Cambridge International AS Level Physics (9702)

250 Extreme-Difficulty Practice Questions

Question Paper

AS syllabus 2025-2027 | Target series: Oct / Nov 2026 | Topics 1–11

Diagrams included · 150 structured (theory) + 100 multiple-choice

Take g = 9.81 m s^{-2} and c = 3.00 × 10^8 m s^{-1} unless stated otherwise.

Reference sheet — what you actually need (AS, topics 1–11)

Data & useful constants

Quantity	Value	Quantity	Value
acceleration of free fall, *g*	9.81 m s^{-2}	speed of light in vacuum, *c*	3.00×10^{8} m s^{-1}
elementary charge, *e*	1.60×10^{-19} C	unified atomic mass unit, *u*	1.66×10^{-27} kg
electron mass	9.11×10^{-31} kg	proton mass	1.67×10^{-27} kg

SI base units: mass (kg), length (m), time (s), current (A), temperature (K). Every equation must be *homogeneous* — base units on both sides match. **Prefixes**: p 10^{-12}, n 10^{-9}, μ 10^{-6}, m 10^{-3}, c 10^{-2}, d 10^{-1}, k 10^{3}, M 10^{6}, G 10^{9}, T 10^{12}. **Uncertainties**: for $y = a \pm b$ add *absolute* uncertainties; for products/quotients ($y = a^m b^n$) add *percentage* uncertainties (× the power). Precision = spread of repeats (random); accuracy = closeness to true value (systematic/zero errors).

Topic	Key relationships
Kinematics	$v = u + at$ · $s = ut + ½at^2$ · $v^2 = u^2 + 2as$ · $s = ½(u+v)t$. Area under *v–t* = displacement; gradient of *v–t* = acceleration; gradient of *s–t* = velocity. Projectiles: independent horizontal (constant v) and vertical (a = g) motions.
Dynamics	$F = ma$ (constant mass) = rate of change of momentum, $F = Δp/Δt$. Momentum $p = mv$. Weight $W = mg$. Conservation of momentum in all collisions; elastic ⇒ KE conserved and relative speed of approach = relative speed of separation. Terminal velocity when drag = weight.
Forces	Moment = $F \times d$ (perp. distance). Torque of a couple = $F \times d$ (separation). Principle of moments: ΣCW = ΣACW about any point. Density $ρ = m/V$; pressure $p = F/A$; hydrostatic $Δp = ρgΔh$; upthrust $F = ρgV$.
Work / energy / power	$W = Fs \cos θ$; $ΔE_P = mgΔh$; $E_K = ½mv^2$; efficiency = useful out / total in; $P = W/t = Fv$.
Deformation	Hooke: $F = kx$. Stress $σ = F/A$, strain $ε = x/L$, Young modulus $E = σ/ε$. Elastic PE = area under *F–x* = $½Fx = ½kx^2$.
Waves	$v = fλ$; $T = 1/f$; intensity = power/area, $I ∈ A^2$. Doppler (moving source): $f_o = f_s v / (v \pm v_s)$. Malus: $I = I_0 \cos^2 θ$. Visible 400–700 nm; all EM waves transverse, speed c.
Superposition	Two-source: $λ = ax/D$. Grating: $d \sin θ = nλ$. Stationary waves: nodes/antinodes, adjacent nodes $λ/2$ apart. Coherence = constant phase difference.
Electricity	$Q = It$; $I = Anvq$; $V = W/Q$; $R = V/I$; $P = VI = I^2R = V^2/R$; $R = ρL/A$.
D.C. circuits	e.m.f. $ε = I(R + r)$; terminal p.d. $V = ε - Ir$. Series $R = R_1+R_2+…$; parallel $1/R = 1/R_1+1/R_2+…$. Kirchhoff I (charge), II (energy). Potential divider $V_{out} = V R_2/(R_1+R_2)$.
Particle physics	Nuclide $^{A}_{Z}X$; A & charge conserved. α = $^{4}_{2}He$, $β^-$ = electron + antineutrino, $β^+$ = positron + neutrino, γ = photon. Quarks: u,d,s,c,t,b (u,c,t = +⅔e; d,s,b = -⅓e). p = uud, n = udd. Baryon = 3 quarks, meson = q&qbar;. Leptons (e, ν) are fundamental.

$β^-$: d → u (n → p). $β^+$: u → d (p → n). α-particles: discrete energies; β-particles: continuous spectrum (energy shared with the (anti)neutrino).

SECTION A — Structured (theory) questions [Q1–Q150]

1 Physical quantities and units

Q1. The viscous drag on a small sphere of radius r moving slowly at speed v through a fluid of viscosity η is $F = 6\pi\eta r v$. **[6]**

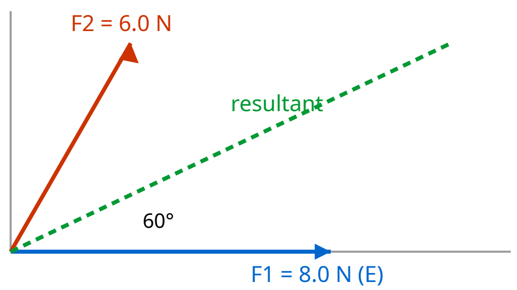
1. Determine the SI base units of η.
2. A student claims the terminal speed of a sphere falling in the fluid is $v = 2r^2(\rho_S-\rho_f)g / 9\eta$, where the ρ are densities. By checking homogeneity, state whether this equation is dimensionally consistent.

Q2. Estimate, showing your reasoning and stating any assumptions, the average useful power developed by an adult who runs up a flight of stairs of typical height in a typical time. Give your answer to one significant figure and comment on why more figures would be unjustified. **[5]**

Q3. A resistance is found from $R = V/I$. In an experiment $V = (6.0 \pm 0.1)$ V and $I = (0.40 \pm 0.02)$ A. **[6]**

1. Calculate R and its absolute uncertainty.
2. The power is $P = V^2/R$. Determine the percentage uncertainty in P.

Q4. Two forces act at a point: F_1 = 8.0 N due east and F_2 = 6.0 N in a direction 60° north of east. By resolving into components, determine the magnitude and direction of the resultant force. **[5]**



Q5. An aircraft has a velocity relative to the air of 250 m s^{-1} due north. A wind blows from the west (i.e. towards the east) at 40 m s^{-1}. **[7]**

1. Find the velocity of the aircraft relative to the ground (magnitude and direction).
2. State the direction in which the pilot must instead head, relative to north, so that the aircraft actually travels due north over the ground, and find the resulting ground speed.

Q6. Distinguish between **random** and **systematic** errors. A micrometer with a zero error of +0.02 mm (it reads 0.02 mm when fully closed) is used to measure the diameter of a wire; five readings are 0.52, 0.53, 0.51, 0.54, 0.52 mm. **[7]**

1. State the type of error caused by the zero error and its effect on the mean.
2. Determine the best estimate of the true diameter and its absolute uncertainty.

Q7. The Young modulus E of a wire is found from $E = 4FL / (\pi d^2 e)$. Measured percentage uncertainties are: F 2%, L 0.5%, d 1.5%, extension e 4%. Determine the percentage uncertainty in E and explain which single measurement it would be most worthwhile to improve. **[5]**

Q8. State what is meant by a **scalar** and a **vector** quantity. For each of the following, state whether it is a scalar or a vector: (i) work done, (ii) momentum, (iii) electric current, (iv) gravitational field strength, (v) power. Give a reason for classifying electric current as you do. **[6]**

Q9. A ball is thrown at a wall and rebounds. Just before impact its velocity is 12 m s^{-1} directed 30° below the horizontal; just after, it is 9.0 m s^{-1} directed 40° above the horizontal. Treating the horizontal (towards the wall) as positive x and upward as positive y, find the magnitude and direction of the **change in velocity** Δv. **[7]**

Q10. The pressure at depth in a liquid is given by $p = h\rho g$. **[5]**

1. Show that this equation is homogeneous.
2. Hence deduce the base units of pressure and confirm they are equivalent to the pascal expressed in terms of the newton.

Q11. A student measures the period T of a pendulum by timing 20 oscillations with a stopwatch of resolution 0.01 s, obtaining a total time of 18.4 s. The reaction time in starting/stopping introduces an uncertainty of about ±0.3 s in the total time. **[6]**

1. Explain why timing 20 oscillations rather than 1 is good practice.
2. Determine T and its percentage uncertainty.

Q12. Estimate the order of magnitude of the number of air molecules in a typical classroom. State your assumptions clearly. (Take the number of molecules per mole as 6×10^{23} and the molar volume of a gas as about 0.024 m^3 mol^{-1} at room conditions.) **[5]**

Q13. Three coplanar forces act on a point object which is in equilibrium: a known force of 10.0 N acting horizontally, and two unknown forces P and Q. P acts at 90° to the 10.0 N force (vertically) and Q acts at 150° to the 10.0 N force (measured anticlockwise). Using the fact that the vector sum is zero, find P and Q. **[7]**

Q14. The drag force on a car at speed v is modelled by $F = bv^2$. **[5]**

1. Determine the SI base units of the constant b.
2. The power needed to overcome this drag is $P = Fv$. Show, using units, that $P = bv^3$ is homogeneous.

Q15. A quantity is calculated from $g = 4\pi^2 L/T^2$ in a pendulum experiment. $L = (1.000 \pm 0.002)$ m and the period is measured as $T = (2.008 \pm 0.005)$ s. **[7]**

1. Calculate g.
2. Determine the absolute uncertainty in g and state the result to an appropriate number of significant figures.

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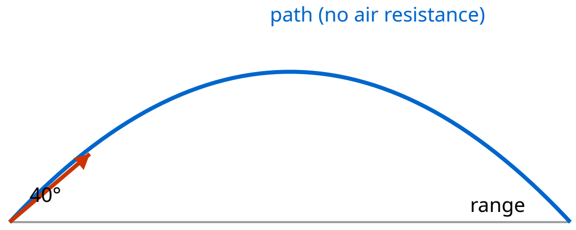
2 Kinematics

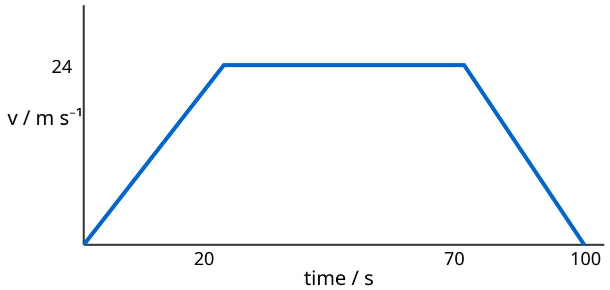
Q16. A ball is projected from the ground with speed 25 m s^{-1} at 40° above the horizontal. Air resistance is negligible. **[8]**

1. Calculate the maximum height reached.
2. Calculate the horizontal range.
3. Determine the speed and direction of the ball 2.5 s after projection.



Q17. The graph shows how the velocity of a train varies with time over a 100 s journey between two stations. Using the graph: **[8]**

1. describe the motion in each of the three phases;
2. determine the total distance travelled between the stations;
3. calculate the magnitude of the (constant) deceleration in the final phase.



Q18. A stone is dropped from rest at the top of a cliff. During the last second before it hits the sea, it falls a distance equal to one quarter of the total height of the cliff. Ignoring air resistance, determine the height of the cliff. (Take g = 9.81 m s^{-2}.) **[7]**

Q19. Describe an experiment, using a falling object and appropriate apparatus, to determine the acceleration of free fall g. Include: the measurements taken, how they are used, one significant source of error, and how it is reduced. **[7]**

Q20. Two cars, A and B, travel along the same straight road. At t = 0 both are level. A moves at a constant 30 m s^{-1}. B starts from rest at t = 0 with constant acceleration 4.0 m s^{-2}. **[8]**

1. Find the time at which B catches up with A.
2. Find the distance travelled by then.
3. Find B's speed at that instant and comment on why it exceeds A's speed.

Q21. A displacement–time graph for an object moving in a straight line is a smooth curve. State how you would obtain, from such a graph, (i) the instantaneous velocity at a point and (ii) the instantaneous acceleration. Explain why the second is harder to obtain accurately. **[6]**

Q22. A projectile is launched horizontally from the top of a tower of height 45 m with speed 18 m s^{-1}. **[7]**

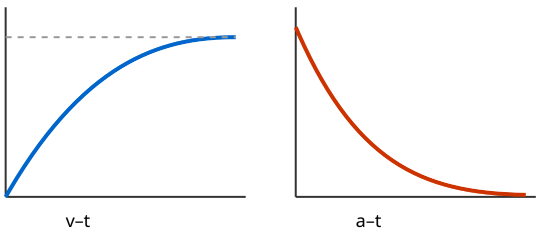
1. Calculate the time to reach the ground.
2. Calculate the horizontal distance from the base of the tower where it lands.
3. Determine the angle the velocity makes with the horizontal on landing.

Q23. Explain, in terms of the independence of horizontal and vertical motion, why a bullet fired horizontally and a bullet dropped from the same height at the same instant reach the ground at the same time (ignoring air resistance and Earth's curvature). State what would change if air resistance were significant. **[6]**

Q24. A car accelerates uniformly from 8.0 m s^{-1} to 20 m s^{-1} while travelling 168 m. **[8]**

1. Calculate the acceleration.
2. Calculate the time taken.
3. Calculate the distance travelled during the final 2.0 s of this period.

Q25. The velocity–time graph shown consists of a curve whose gradient decreases with time, levelling off to a horizontal asymptote. State what kind of motion this represents, sketch the corresponding acceleration–time graph, and identify a real physical situation the graph could describe. **[6]**



Q26. A ball is thrown vertically upward from a height of 1.5 m above the ground with an initial speed of 12 m s^{-1}. Taking upward as positive and g = 9.81 m s^{-2}: **[8]**

1. find the total time before it strikes the ground;
2. find its speed just before impact;
3. sketch (with values) the velocity–time graph for the whole motion.

Q27. An object moves so that its displacement is given by successive equal time intervals of 1.0 s: the distances travelled in the 1st, 2nd, 3rd and 4th seconds are 5, 15, 25 and 35 m respectively. **[7]**

1. Show that the motion is uniformly accelerated and find the acceleration.
2. Find the initial velocity.

Q28. A cyclist travels the first half of a straight journey at 6.0 m s^{-1} and the second half (equal distance) at 12 m s^{-1}. Later the cyclist repeats the journey travelling at 6.0 m s^{-1} for the first half of the *time* and 12 m s^{-1} for the second half of the time. **[8]**

1. Find the average speed for the equal-distance journey.
2. Find the average speed for the equal-time journey.
3. Explain which is larger and why.

Q29. During a tennis serve the ball is struck at a height of 2.4 m and travels horizontally (initially) at 40 m s^{-1} towards the net 12 m away. The net is 0.91 m high. Ignoring air resistance and any vertical launch component, determine whether the ball clears the net, and by what margin. **[7]**

Q30. A lift (elevator) starts from rest, accelerates upward at 1.2 m s^{-2} for 3.0 s, then moves at constant velocity for 5.0 s, then decelerates uniformly to rest in a further 2.0 s. **[8]**

1. Sketch the velocity–time graph.
2. Find the total distance travelled.
3. Find the average velocity for the whole journey.

Q31. A particle is projected up a smooth plane inclined at 25° to the horizontal with an initial speed of 9.0 m s^{-1} along the plane. The component of gravitational acceleration along the plane is g sin25°. **[7]**

1. Find the distance travelled up the plane before it momentarily stops.
2. Find the time taken to return to its starting point.

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3 Dynamics

Q32. State Newton's second law in terms of momentum. A jet of water from a hose leaves a nozzle of cross-sectional area 4.0×10^{-4} m² at a speed of 25 m s^{-1} and strikes a vertical wall at right angles, running down the wall without rebounding. (Density of water = 1.0×10^3 kg m^{-3}.) **[7]**

1. Calculate the mass of water hitting the wall per second.
2. Calculate the force exerted on the wall.

Q33. A trolley of mass 0.80 kg moving at 3.0 m s^{-1} collides head-on with a stationary trolley of mass 1.2 kg. After the collision they move off separately; the 0.80 kg trolley continues in the same direction at 0.60 m s^{-1}. **[8]**

1. Find the velocity of the 1.2 kg trolley after the collision.
2. Determine whether the collision is elastic, showing your working.

Q34. A ball of mass 0.15 kg strikes a wall horizontally at 8.0 m s^{-1} and rebounds horizontally at 6.0 m s^{-1}. The contact time is 0.020 s. **[7]**

1. Calculate the magnitude of the change in momentum of the ball.
2. Calculate the average force exerted by the wall on the ball.
3. State the force exerted by the ball on the wall and justify it.

Q35. Explain, using Newton's laws, why a person of weight 700 N standing in a lift experiences a different reading on bathroom scales placed under their feet when the lift (a) accelerates upward at 2.0 m s^{-2} and (b) accelerates downward at 2.0 m s^{-2}. Calculate the scale readings. **[8]**

Q36. A rocket works by ejecting exhaust gas. In a test, a rocket ejects gas at a rate of 45 kg s^{-1} with a speed of 1200 m s^{-1} relative to the rocket. **[7]**

1. Calculate the thrust produced.
2. If the rocket has an initial total mass of 6.0×10^3 kg, find its initial acceleration at lift-off (vertically upward).

Q37. Two ice skaters, of masses 60 kg and 45 kg, stand at rest facing each other and push apart. The 60 kg skater moves off at 1.5 m s^{-1}. **[7]**

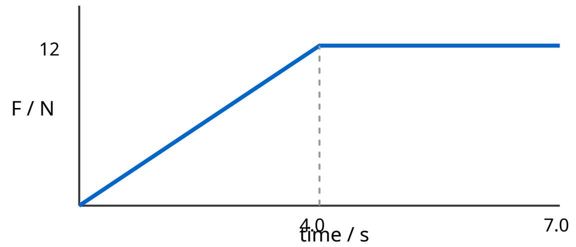
1. Find the speed of the 45 kg skater.
2. Find the total kinetic energy generated and state where it came from.
3. State the total momentum of the system after the push and explain your answer.

Q38. A ball A of mass m moving at speed u makes a one-dimensional elastic collision with a stationary ball B of mass 3m. Using conservation of momentum and the elastic-collision condition, determine the velocities of A and B after the collision in terms of u, and comment on the direction of A's motion. **[8]**

Q39. Define linear momentum and state the principle of conservation of momentum, including the condition under which it applies. Explain how this principle is consistent with Newton's third law for two interacting bodies. **[6]**

Q40. An object of mass 2.0 kg experiences a resultant force that varies with time as shown: the force rises linearly from 0 to 12 N over the first 4.0 s, then stays constant at 12 N for the next 3.0 s. The object starts from rest. **[7]**

1. Use the area under the force–time graph to find the impulse over the 7.0 s.
2. Hence find the final velocity.



Q41. A raindrop falling through still air reaches a terminal velocity. **[8]**

1. Explain, in terms of the forces acting, how terminal velocity is reached.
2. Sketch and describe the acceleration–time graph from the instant the drop starts to fall.
3. State and explain what happens to the terminal velocity if the drop coalesces with others to form a larger drop.

Q42. A 1500 kg car travelling at 20 m s^{-1} collides with and sticks to a stationary 900 kg car (a perfectly inelastic collision). The combined wreck then slides to rest on a road for which the constant resistive (frictional) force is 8.4×10^3 N. **[9]**

1. Find the common velocity immediately after impact.
2. Find the distance the wreck slides before stopping.
3. Find the kinetic energy lost in the collision itself.

Q43. A particle of mass 2.0 kg moving at 4.0 m s^{-1} along the +x direction collides with a 3.0 kg particle at rest. After the collision the 2.0 kg particle moves at 2.0 m s^{-1} at 60° above the +x axis. Using conservation of momentum in two dimensions, find the velocity (magnitude and direction) of the 3.0 kg particle after the collision. **[9]**

Q44. A constant horizontal force of 50 N is applied to a 12 kg box on a rough horizontal floor. The box accelerates at 2.5 m s^{-2}. **[8]**

1. Find the frictional force acting on the box.
2. The force is then removed. Find the deceleration of the box, assuming the frictional force is unchanged, and the distance it travels before stopping if it was moving at 6.0 m s^{-1} when the force was removed.

Q45. Explain the distinction between mass and weight. A probe of mass 250 kg is taken to a planet where the gravitational field strength is 3.7 N kg^{-1}. **[7]**

1. State its mass and calculate its weight on the planet.
2. Calculate the resultant force and acceleration if a 1200 N thrust acts vertically upward on the probe at the planet's surface.

Q46. A chain of total mass 3.0 kg and length 2.0 m hangs vertically with its lower end just touching a table. It is released and falls; as it lands, the links come to rest on the table. Consider the instant when a length x of chain has already landed. **[6]**

1. Explain why the force the table exerts on the chain is greater than the weight of the chain already at rest on it.
2. Outline the two contributions to this force.

Q47. State Newton's first law. A book rests on a table. Identify the forces on the book, state which pairs are Newton's-third-law pairs, and explain why the upward push of the table on the book and the weight of the book are **not** a Newton's-third-law pair even though they are equal and opposite here. **[6]**

Q48. Two gliders on a frictionless air track approach each other: glider P (0.30 kg) moves right at 0.80 m s^{-1}; glider Q (0.50 kg) moves left at 0.40 m s^{-1}. They collide and the collision is perfectly elastic. Find the velocity of each glider after the collision. **[9]**

Q49. A fireworks shell of mass 2.0 kg is moving vertically upward at 15 m s^{-1} when it explodes into two fragments. One fragment of mass 0.80 kg is thrown vertically downward at 12 m s^{-1} immediately after the explosion. **[7]**

1. Find the velocity of the other fragment immediately after the explosion.
2. State and justify the total momentum immediately before and after the explosion.

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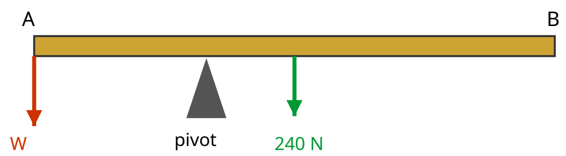
4 Forces, density and pressure

Q50. A uniform beam AB of weight 240 N and length 3.0 m is pivoted at a point 1.0 m from end A. A load of weight W hangs from end A, and a downward force is applied at end B to keep the beam horizontal and in equilibrium. **[7]**

1. Taking moments about the pivot, find the relationship needed for equilibrium if no force acts at B and only W balances the beam's weight.
2. Hence find the value of W that keeps the beam horizontal with no force at B.



Q51. Define the **moment of a force** and the **torque of a couple**. A steering wheel of diameter 0.36 m is turned by a driver applying two equal, oppositely-directed tangential forces of 15 N at opposite ends of a diameter. **[6]**

1. Calculate the torque of the couple.
2. Explain why a couple produces rotation but no translational acceleration.

Q52. A rectangular block of dimensions 0.20 m × 0.15 m × 0.10 m has a mass of 8.1 kg. **[7]**

1. Calculate its density.
2. Calculate the maximum and minimum pressures it can exert when resting on a horizontal surface on different faces.

Q53. Starting from the definitions of pressure and density, derive the expression $\Delta p = \rho g \Delta h$ for the pressure difference between two points separated by a vertical height Δh in a fluid of uniform density ρ. **[6]**

Q54. A U-tube contains water. Oil of density 800 kg m^{-3} is poured into the left arm and floats on the water without mixing. The oil column is 12 cm long. When equilibrium is reached, the water level in the right arm is higher than the oil–water boundary in the left arm. **[7]**

1. Explain why there is a height difference.
2. Calculate the difference in the levels of the two liquid surfaces (density of water = 1000 kg m^{-3}).

Q55. An object of volume 5.0×10^{-4} m^3 and weight 6.0 N is fully submerged and held stationary in water (density 1000 kg m^{-3}). **[8]**

1. Calculate the upthrust on the object.
2. Determine the tension in the string holding it, and state whether the string is above or below the object.
3. Explain the origin of the upthrust in terms of pressure.

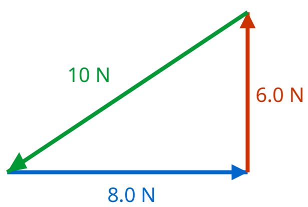
Q56. A non-uniform plank of length 4.0 m rests horizontally on two supports, one at each end. The support at end A reads 320 N and the support at end B reads 480 N. **[7]**

1. Find the weight of the plank.
2. Find the distance of the centre of gravity from end A.

Q57. State the two conditions for a rigid body to be in equilibrium. A ladder leans against a smooth vertical wall, resting on rough ground. Explain, without calculation, why the frictional force at the ground is essential for equilibrium and in which direction it acts. **[6]**

Q58. Three coplanar forces act on a body in equilibrium. Forces of 6.0 N and 8.0 N act at right angles to each other. **[6]**

1. Use a vector triangle to find the magnitude and direction of the third force.
2. State the general rule that three forces in equilibrium must satisfy when drawn head-to-tail.



Q59. A hydraulic press has a small piston of area 2.0×10^{-4} m^2 and a large piston of area 5.0×10^{-2} m^2. A force of 40 N is applied to the small piston. **[8]**

1. Using the fact that pressure is transmitted equally through the fluid, find the force on the large piston.
2. The small piston moves down 0.30 m. Find the distance the large piston rises, and show energy is conserved.

Q60. A uniform rod of weight 50 N and length 1.2 m is hinged at a wall at end A and held horizontal by a cable attached at end B, making an angle of 30° with the rod. **[7]**

1. By taking moments about A, find the tension in the cable.
2. Hence state one advantage of taking moments about A rather than about the centre.

Q61. A helium balloon of volume 0.40 m^3 is filled with helium of density 0.18 kg m^{-3}. The mass of the (empty) balloon fabric and its attachments is 0.050 kg. The surrounding air has density 1.2 kg m^{-3}. **[8]**

1. Calculate the upthrust on the balloon.
2. Determine the maximum additional load the balloon can just lift.

Q62. Explain what is meant by the **centre of gravity** of an object. Describe how you would find the centre of gravity of an irregular flat lamina experimentally, and explain the physics that makes the method work. **[6]**

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5 Work, energy and power

Q63. A cyclist and bicycle of total mass 90 kg free-wheel from rest down a hill that drops a vertical height of 40 m over a slope length of 500 m. At the bottom the speed is 22 m s^{-1}. **[8]**

1. Calculate the loss in gravitational potential energy.
2. Calculate the kinetic energy gained.
3. Hence find the average resistive force acting along the slope.

Q64. Derive the expression P = Fv for the power delivered by a constant force F acting on an object moving at constant velocity v in the direction of the force. A car engine delivers a useful power of 24 kW to drive a car at a steady 30 m s^{-1} on a level road. **[8]**

1. Calculate the total resistive force at this speed.
2. If the resistive force is proportional to v^2, find the useful power needed to travel at 40 m s^{-1}.

Q65. A pump raises water from a well 8.0 m deep and delivers it at the surface through a pipe at a speed of 3.0 m s^{-1}. The mass of water delivered per second is 15 kg. **[8]**

1. Calculate the power needed to raise the water (gain in gravitational PE per second).
2. Calculate the power needed to give the water its kinetic energy.
3. If the pump's input power is 2.0 kW, calculate its efficiency.

Q66. A ball of mass 0.20 kg is dropped from a height of 2.0 m onto a hard floor and rebounds to a height of 1.4 m. **[8]**

1. Calculate the speed of the ball just before and just after the bounce.
2. Calculate the energy lost in the bounce and express it as a percentage of the initial PE.
3. State the main form into which the lost energy is transferred.

Q67. A skier of mass 70 kg starts from rest at the top of a slope and reaches a speed of 18 m s^{-1} at the bottom, having descended a vertical height of 25 m along a track of length 90 m. **[8]**

1. Show that the motion is not frictionless and find the total energy dissipated.
2. Find the average frictional force and the efficiency of the descent in converting PE to KE.

Q68. State the principle of conservation of energy. A 0.050 kg arrow is fired vertically upward and leaves the bow with 30 J of kinetic energy. Air resistance does 6.0 J of work on the arrow during the ascent. **[7]**

1. Find the maximum height reached.
2. Explain why the arrow returns to the ground with less than 30 J of kinetic energy.

Q69. A conveyor belt lifts gravel at a rate of 25 kg s^{-1} onto a platform 4.0 m higher than the loading point. The gravel is placed on the belt essentially at rest and leaves it moving at the belt speed of 1.5 m s^{-1}. **[7]**

1. Calculate the power needed to raise the gravel.
2. Calculate the additional power needed to give the gravel its kinetic energy.
3. State one reason the motor's actual power input must exceed the sum of these.

Q70. A 1200 kg car accelerates on a level road from 10 m s^{-1} to 25 m s^{-1} in 8.0 s. The average resistive force during this time is 500 N. **[9]**

1. Calculate the gain in kinetic energy.
2. Calculate the average driving force provided by the engine.
3. Calculate the average useful output power of the engine over this period.

Q71. A simple pendulum bob of mass 0.30 kg is pulled aside so that it rises a vertical height of 0.20 m above its lowest point, then released. **[8]**

1. Find the speed of the bob at the lowest point (neglecting resistance).
2. The string is 1.0 m long. Find the tension in the string at the lowest point.

Q72. Define **efficiency**. An electric motor rated at 500 W is used to lift a 30 kg load. In a test, it raises the load 6.0 m in 5.0 s at constant speed. **[7]**

1. Calculate the useful power output.
2. Calculate the efficiency.
3. Explain, in energy terms, where the 'lost' input energy goes.

Q73. A spring-loaded toy of mass 0.040 kg is compressed against a spring storing 0.90 J of elastic potential energy, then released to fire vertically upward. Air resistance is negligible. **[7]**

1. Assuming all the elastic PE becomes kinetic energy, find the launch speed.
2. Find the maximum height reached above the launch point.

Q74. Explain why, for a car braking to rest, the braking distance is proportional to the square of the initial speed (assuming a constant braking force). A car travelling at 15 m s^{-1} stops in 18 m. Estimate its stopping distance from 30 m s^{-1} with the same braking force. **[6]**

Q75. A ski-lift carries skiers up a 30° slope at a constant 2.5 m s^{-1}. Each chair plus skier has an average mass of 85 kg and there are 40 loaded chairs on the ascending side at any instant. Friction is negligible. **[8]**

1. Find the component of weight along the slope for one chair.
2. Find the total useful power the motor must supply to the ascending chairs.

Q76. A 0.60 kg block slides along a horizontal frictionless surface at 5.0 m s^{-1}, then encounters a rough patch 2.0 m long where the frictional force is 1.8 N, after which the surface is frictionless again and rises into a smooth ramp. **[7]**

1. Find the block's kinetic energy after crossing the rough patch.
2. Find the maximum vertical height it rises up the ramp.

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6 Deformation of solids

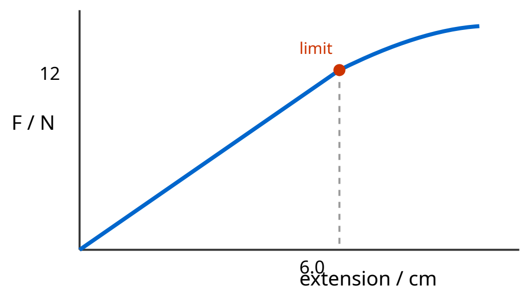
Q77. A steel wire of natural length 2.5 m and diameter 0.40 mm is stretched by a load of 45 N, producing an extension of 2.8 mm. **[8]**

1. Calculate the stress in the wire.
2. Calculate the strain.
3. Hence calculate the Young modulus of steel.

Q78. Describe an experiment to determine the Young modulus of a metal in the form of a long wire. State the measurements taken and the instrument used for each, how the result is obtained, and two precautions that improve accuracy. **[8]**

Q79. The force–extension graph for a spring is a straight line from the origin up to a load of 12 N at an extension of 6.0 cm, beyond which the line curves (the limit of proportionality is reached at 12 N). **[7]**

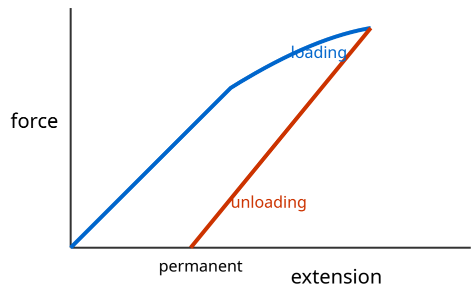
1. Calculate the spring constant.
2. Calculate the elastic potential energy stored at an extension of 6.0 cm.
3. State what the area under the graph beyond the limit of proportionality represents.



Q80. Two wires, X and Y, are made of the same material. Y has twice the diameter and half the length of X. The same load is hung from each. **[7]**

1. Compare the stress in Y with that in X.
2. Compare the extension of Y with that of X.

Q81. Distinguish between **elastic** and **plastic** deformation, and define the **elastic limit**. A metal wire is loaded beyond its elastic limit and then completely unloaded. Sketch and describe the loading and unloading force–extension lines, and identify what the area between them represents. **[7]**



Q82. A vertical spring of spring constant 25 N m^{-1} hangs with a 0.15 kg mass attached at rest. **[8]**

1. Find the extension of the spring.
2. The mass is pulled down a further 4.0 cm and released. Using energy methods, find the elastic PE stored at the moment of release relative to the equilibrium position, and hence the maximum speed of the mass as it passes back through equilibrium.

Q83. Define **stress**, **strain** and the **Young modulus**, giving the SI unit of each. Explain why the Young modulus is a property of the material, whereas the spring constant is a property of the particular specimen. **[6]**

Q84. A composite hangs from a support: a steel wire of stiffness (force constant) 800 N m^{-1} is joined end-to-end (in series) to a second wire of stiffness 1200 N m^{-1}. A load of 24 N is hung from the lower end. **[7]**

1. Explain why the two wires experience the same force.
2. Find the total extension of the combination.

Q85. The graph of stress against strain for a ductile metal wire is drawn. Mark and explain the significance of (i) the region where the graph is a straight line through the origin, (ii) the point where the graph first departs from a straight line, and (iii) the region of large strain for small increases in stress. **[6]**

Q86. A rubber cord and a metal wire are each stretched and then released. State and explain, with reference to their force–extension behaviour, one important way in which rubber differs from the metal in how it stores and returns energy. **[5]**

Q87. A wire obeys Hooke's law up to a load of 60 N, at which its extension is 4.0 mm. **[7]**

1. Calculate the work done in stretching it to this extension.
2. An identical wire is stretched to only 3.0 mm. Calculate the elastic PE stored, and state what fraction of the answer to (a) this represents.

7 Waves

Q88. Define **displacement**, **amplitude**, **frequency** and **phase difference** for a progressive wave. **[7]**
Two points on a progressive wave of wavelength 0.80 m are separated by 0.30 m along the direction of travel.

1. Calculate their phase difference in degrees and in radians.
2. State the smallest separation of two points that oscillate in antiphase.

Q89. The trace on a cathode-ray oscilloscope (CRO) shows a sinusoidal signal. The time-base is set to 2.0 ms per division and one complete cycle occupies 3.5 divisions horizontally. The y-gain is 0.50 V per division and the trace has a peak-to-peak height of 6.0 divisions. **[6]**

1. Calculate the frequency of the signal.
2. Calculate the amplitude (peak voltage) of the signal.

Q90. Derive the wave equation $v = f\lambda$ from the definitions of speed, frequency and wavelength. A sound wave of frequency 256 Hz travels in air at 340 m s^{-1} and then passes into water where its speed becomes 1480 m s^{-1}. **[7]**

1. Find its wavelength in air and in water.
2. State, with a reason, what happens to the frequency as the wave crosses into the water.

Q91. A point source emits sound uniformly in all directions with a power of 0.80 W. **[7]**

1. Calculate the intensity at a distance of 5.0 m from the source.
2. At what distance is the intensity one quarter of this value?
3. State how the amplitude of the wave at that greater distance compares with the amplitude at 5.0 m.

Q92. Compare **transverse** and **longitudinal** waves, giving one example of each from the syllabus. **[6]**
Explain, in terms of the direction of particle oscillation, why one of these wave types can be polarised and the other cannot.

Q93. A police car siren emits sound of frequency 900 Hz. The car moves at 30 m s^{-1} along a straight road; the speed of sound is 340 m s^{-1}. **[8]**

1. Calculate the frequency heard by a stationary observer as the car approaches.
2. Calculate the frequency heard as the car recedes.
3. Explain physically why the approaching frequency is higher.

Q94. State the approximate range of wavelengths (in free space) of the following regions of the electromagnetic spectrum and place them in order of increasing wavelength: X-rays, microwaves, visible light, infrared. State two properties that all electromagnetic waves share. **[6]**

Q95. Unpolarised light of intensity I_0 passes through a polarising filter, then through a second (analyser) whose transmission axis is at 60° to the first. **[7]**

1. State the intensity of the light after the first filter (in terms of I_0).
2. Use Malus's law to find the intensity after the second filter.
3. State the effect on the final intensity of rotating the analyser to 90° to the first.

Q96. A wave travelling in the +x direction is described by a displacement–position graph (a 'snapshot' at one instant) and, separately, by a displacement–time graph for one particle. State what physical quantity is read directly from each graph, and explain how the two graphs differ even though both look sinusoidal. **[6]**

Q97. A loudspeaker connected to a signal generator produces a sound of constant frequency. As the output power of the generator is doubled, describe and explain the change (if any) in (i) the frequency of the sound, (ii) the wavelength, and (iii) the amplitude of the air-particle oscillations, assuming the speed of sound is constant. **[6]**

Q98. The speed of a transverse wave on a stretched string depends on the tension and the mass per unit length. In an investigation, a wave of frequency 50 Hz on a string has a wavelength of 1.2 m. **[6]**

1. Calculate the wave speed.
2. The tension is then increased so the speed becomes 90 m s^{-1} while the frequency is kept at 50 Hz. Find the new wavelength.

Q99. Explain what is meant by saying that a progressive wave **transfers energy** without transferring matter. Use the example of a cork floating on water as a water wave passes to support your explanation. **[5]**

Q100. A beam of vertically plane-polarised light of intensity I_0 is incident on a series of three polarising filters whose transmission axes are at 0°, 45° and 90° to the vertical respectively. **[7]**

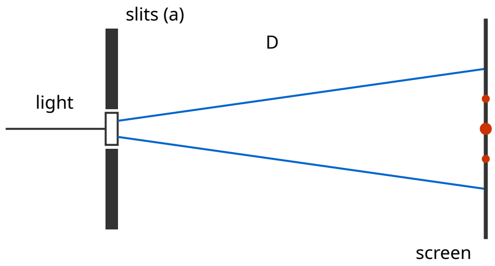
1. Find the intensity transmitted through all three, in terms of I_0.
2. Compare this with the intensity if the middle filter were removed, and comment.

Q101. A microwave source and detector are used to show that microwaves are transverse and to measure their properties. Describe (i) how you would demonstrate that the microwaves are polarised, and (ii) outline how you could estimate the wavelength using a metal reflector. **[6]**

8 Superposition

Q102. State the **principle of superposition**. In a Young's double-slit experiment, coherent light of wavelength 590 nm illuminates two slits 0.45 mm apart, and fringes are observed on a screen 2.4 m away. **[7]**

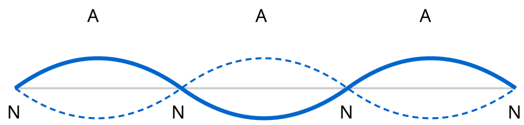
1. Calculate the fringe separation.
2. Calculate the distance from the central bright fringe to the 4th-order bright fringe.



Q103. Explain the meaning of the terms **interference** and **coherence**. State the conditions necessary for a stable two-source interference pattern of light to be observed, and explain why two separate filament lamps of the same colour cannot produce such a pattern. **[7]**

Q104. A stationary (standing) wave is set up on a string of length 1.5 m fixed at both ends, vibrating in its third harmonic (three loops). **[7]**

1. Sketch the string, marking nodes (N) and antinodes (A).
2. Determine the wavelength.
3. If the wave speed on the string is 120 m s^{-1}, find the frequency.



Q105. A diffraction grating has 500 lines per millimetre. It is illuminated normally by monochromatic light of wavelength 640 nm. **[8]**

1. Calculate the grating spacing d.
2. Calculate the angle of the second-order maximum.
3. Determine the highest order that can be observed.

Q106. Explain the formation of a stationary wave on a stretched string in terms of two progressive waves. State two ways in which a stationary wave differs from a progressive wave. **[7]**

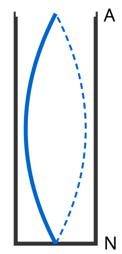
Q107. Explain the meaning of **diffraction**. Describe, using a ripple tank, how the extent of diffraction of water waves through a gap depends on the size of the gap relative to the wavelength, and state the condition for the most pronounced diffraction. **[6]**

Q108. In a double-slit experiment the fringe separation is measured for red light and then for blue light with the same slits and screen distance. **[6]**

1. State and explain which colour gives the wider fringe spacing.
2. The screen is then moved twice as far away. State the effect on the fringe separation.
3. Instead, the slit separation is doubled. State the effect on the fringe separation.

Q109. A tube closed at one end and open at the other resonates with sound. The shortest air column that gives resonance with a tuning fork of frequency 512 Hz is 16.2 cm long. (Speed of sound in air = 330 m s^{-1}; ignore end corrections.) **[7]**

1. Sketch the displacement pattern for this fundamental resonance, marking the node and antinode.
2. Show that the data are consistent with the speed of sound quoted.



Q110. Two identical loudspeakers, driven in phase by the same signal generator at a frequency of 1.7 kHz, are placed 1.5 m apart facing a listener who walks along a line parallel to the speakers. The listener notices alternate loud and quiet points. (Speed of sound = 340 m s^{-1}.) **[7]**

1. Explain the origin of the loud and quiet points.
2. State the path difference (in wavelengths) at a quiet point nearest the centre.

Q111. Describe how a diffraction grating can be used to measure the wavelength of monochromatic light. State the measurements taken and how the wavelength is calculated, and explain one advantage of using a grating rather than a double slit for this purpose. **[7]**

Q112. White light is passed through a diffraction grating. **[6]**

1. Explain why the central (zero-order) maximum is white but the higher orders are spread into spectra.
2. State, with a reason, which colour is diffracted through the largest angle in a given order.

Q113. Microwaves of wavelength 2.8 cm from a transmitter are reflected back by a metal sheet, forming a stationary wave. A small probe detector is moved along the line between transmitter and sheet. **[6]**

1. Explain what the detector registers as it is moved, and why.
2. Calculate the distance the detector moves between five successive positions of minimum signal.

Q114. In a Young's double-slit arrangement, the fringes are found to have low contrast (bright fringes not very bright, dark fringes not fully dark). Suggest two possible reasons for the poor contrast and, for each, state how it could be improved. **[6]**

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9 Electricity

Q115. A copper wire of cross-sectional area 1.5×10^{-6} m² carries a current of 3.2 A. Copper has 8.5×10^{28} free electrons per cubic metre. **[7]**

1. Calculate the drift speed of the electrons.
2. Comment on the size of your answer, given that electrical signals travel near the speed of light.

Q116. State what is meant by 'the charge on charge carriers is quantised'. A charge of 96 C flows past a point in a conductor in 40 s. **[6]**

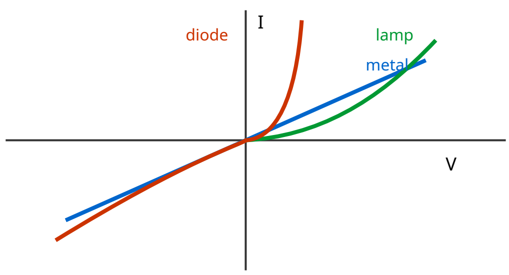
1. Calculate the current.
2. Calculate the number of electrons that pass the point in this time.

Q117. Define **potential difference** across a component. A 12 V battery drives a current of 0.50 A through a motor for 2.0 minutes, lifting a load. **[8]**

1. Calculate the charge that flows.
2. Calculate the electrical energy transferred.
3. If the motor lifts a 1.5 kg load by 40 cm in this time, find its efficiency.

Q118. The I–V characteristics of a metallic conductor at constant temperature, a filament lamp, and a semiconductor diode differ markedly. **[8]**

1. Sketch all three characteristics on the same axes.
2. Explain the shape of the filament-lamp characteristic.
3. State how the resistance of the diode varies as the forward voltage increases from zero.



Q119. State **Ohm's law**. A nichrome wire has resistance 6.0 Ω. A second nichrome wire of the same material has twice the length and half the diameter of the first. **[7]**

1. Calculate the resistance of the second wire.
2. Explain, using $R = \rho L/A$, each factor by which the resistance changes.

Q120. A resistivity experiment gives, for a wire of uniform diameter 0.38 mm and length 1.20 m, a resistance of 4.6 Ω. **[8]**

1. Calculate the resistivity of the material.
2. The diameter is measured with a micrometer to ±0.01 mm and the length to ±1 cm. State which measurement contributes most to the percentage uncertainty in the resistivity, and why.

Q121. Define **resistance.** A 60 W, 240 V lamp and a 100 W, 240 V lamp are each operated at their rated values. **[7]**

1. Calculate the operating resistance of each lamp.
2. State, with a reason, which lamp has the thicker or shorter filament (assuming same material and operating temperature).

Q122. Explain, in terms of charge carriers, why the resistance of (i) a thermistor decreases as its temperature rises and (ii) a light-dependent resistor (LDR) decreases as light intensity increases. Contrast this with the behaviour of a metal as its temperature rises. **[7]**

Q123. A 2.5 kW electric heater is designed for a 230 V mains supply. **[7]**

1. Calculate the current it draws and its resistance.
2. The heater is instead connected to a 115 V supply. Assuming its resistance is unchanged, calculate the power now dissipated and comment on the result.

Q124. The current in a conductor is not the same thing as the drift of individual electrons. Using $I = Anvq$, explain what happens to the drift speed when a wire of the same material narrows to half its cross-sectional area, if the current is unchanged. State what this implies about charge conservation. **[6]**

Q125. A student has a length of resistance wire and wants a 5.0 Ω resistor. The wire has resistance 2.0 Ω per metre. **[7]**

1. Find the length needed.
2. The student instead cuts three equal lengths of this same 2.5 m wire and connects them in parallel. Find the resistance of the combination.

Q126. A 100 m length of overhead power cable has a total resistance of 0.40 Ω and carries a current of 250 A. **[6]**

1. Calculate the power dissipated as heat in the cable.
2. Explain, with reference to $P = I^2R$, why electrical power is transmitted at high voltage and low current.

Q127. Two wires P and Q are connected in series across a battery. P has resistance 4.0 Ω and Q has resistance 8.0 Ω. **[6]**

1. Compare the current in P with that in Q.
2. Compare the power dissipated in P with that in Q.
3. Explain your answer to (b) using $P = I^2R$.

10 D.C. circuits

Q128. A cell of e.m.f. 1.5 V and internal resistance 0.50 Ω is connected to an external resistor of 2.5 Ω. **[7]**

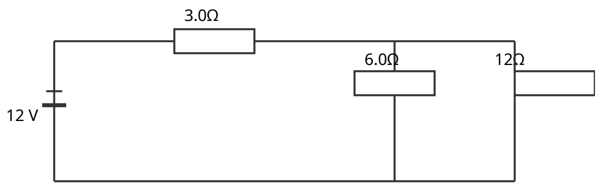
1. Calculate the current in the circuit.
2. Calculate the terminal potential difference.
3. Calculate the power wasted inside the cell.

Q129. State **Kirchhoff's first and second laws**, and state the conservation principle underlying each. Explain briefly why the first law must hold at any junction in a circuit that carries a steady current. **[6]**

Q130. Derive, using Kirchhoff's laws, the formula for the combined resistance of two resistors R_1 and R_2 connected in parallel across a supply of p.d. V. **[6]**

Q131. In the circuit, a 12 V battery of negligible internal resistance is connected to a 3.0 Ω resistor in series with a parallel combination of a 6.0 Ω and a 12 Ω resistor. **[8]**

1. Find the total resistance.
2. Find the current drawn from the battery.
3. Find the current in the 12 Ω resistor.



Q132. Distinguish between the **e.m.f.** of a source and the **potential difference** across its terminals, in terms of energy. Explain why the terminal p.d. falls below the e.m.f. when the source delivers a current, and describe one situation in which the terminal p.d. equals the e.m.f. **[6]**

Q133. A potential divider consists of two fixed resistors, R_1 = 4.0 kΩ (top) and R_2 = 6.0 kΩ (bottom), across a 9.0 V supply, with the output taken across R_2. **[8]**

1. Calculate the output voltage on no load.
2. A voltmeter of resistance 6.0 kΩ is now connected across the output. Calculate the new output reading, and comment.

Q134. Explain the principle of a **potentiometer** used as a means of comparing two potential differences, and describe the role of the galvanometer in the null method. State one advantage of this method over using a voltmeter. **[7]**

Q135. Two identical cells, each of e.m.f. 1.5 V and internal resistance 0.30 Ω, are connected to a 1.4 Ω external resistor. **[8]**

1. Find the current if the cells are connected in series.
2. Find the current if the cells are connected in parallel.

Q136. A thermistor is used in a potential divider to make a temperature-sensing circuit that switches on a warning when the temperature becomes too high. The thermistor (negative temperature coefficient) is placed in series with a fixed resistor across a 6.0 V supply. **[7]**

1. State how the thermistor's resistance changes as temperature rises.
2. Explain how you would connect the output so that the output voltage **rises** as the temperature rises, and justify your choice.

Q137. In the circuit shown, a 6.0 V battery of internal resistance 0.50 Ω supplies two resistors: R_1 = 10 Ω in parallel with R_2 = 15 Ω. Use Kirchhoff's laws to find the terminal p.d. and the current in each resistor. **[8]**

Q138. A battery of e.m.f. ε and internal resistance r delivers current to a variable external resistance R. Explain how the terminal p.d. and the power delivered to R vary as R is decreased from a very large value towards zero, and state the condition for maximum power transfer to R. **[7]**

Q139. A student connects a 12 V battery (internal resistance 1.0 Ω) to a network: a 4.0 Ω resistor in series with two 6.0 Ω resistors that are in parallel with each other. **[9]**

1. Find the current drawn from the battery.
2. Find the power dissipated in the 4.0 Ω resistor.
3. Find the total power delivered by the battery and the fraction wasted internally.

Q140. Describe how a light-dependent resistor (LDR) can be used in a potential divider to switch on a lamp automatically when it gets dark. Draw/describe the arrangement and explain the operation. **[6]**

Q141. Two resistors of 200 Ω and 300 Ω are available. A 5.0 V supply of negligible internal resistance is connected across them. **[7]**

1. Find the total power dissipated when they are in series and when they are in parallel.
2. State which arrangement dissipates more power and explain why using $P = V^2/R$.

11 Particle physics

Q142. In the α-particle scattering experiment, a beam of α-particles is directed at a thin gold foil. Most pass almost straight through, a small fraction are deflected through large angles, and a very few are reflected back. **[7]**

1. State what each of these three observations tells us about the atom.
2. Explain why a thin foil and a vacuum are used.

Q143. A nuclide is written as $^{A}_{Z}X$. **[7]**

1. State what A and Z represent.
2. The nuclide $^{238}_{92}U$ decays by α-emission to thorium (Th). Write the balanced decay equation and state the values of A and Z for the thorium nuclide.
3. Explain what is conserved in the decay.

Q144. Describe the composition, relative mass and relative charge of α-, β^-- and γ-radiations. Explain why α-particles from a given decay have discrete energies whereas β-particles have a continuous range of energies. **[8]**

Q145. State what is meant by an **antiparticle**. Write the nuclear equation for the β^+ decay of a nucleus $^{A}_{Z}X$, naming all particles produced, and state the change (if any) in A and Z. **[7]**

Q146. Quarks are fundamental particles. **[7]**

1. State the charges (in units of e) of the up, down and strange quarks.
2. Show that the quark composition of a proton (uud) and a neutron (udd) gives the correct charges.
3. State the composition of a baryon and of a meson in terms of quarks.

Q147. Describe, in terms of quarks, the changes that occur during (i) β^- decay and (ii) β^+ decay. State which fundamental particles (leptons) are emitted in each. **[7]**

Q148. The unified atomic mass unit (u) is used as a unit of mass in nuclear physics. **[6]**

1. State the approximate value of 1 u in kilograms.
2. A carbon-12 nucleus has 6 protons and 6 neutrons. Using masses of 1.007 u (proton) and 1.009 u (neutron), calculate the total mass of the separate nucleons in u, and comment on why this exceeds 12 u.

Q149. Classify each of the following as a lepton, a baryon or a meson, giving a brief reason: (i) a proton, (ii) an electron, (iii) a particle consisting of an up quark and a down antiquark, (iv) a neutrino. **[6]**

Q150. A nucleus $^{14}_{6}C$ undergoes β^- decay to nitrogen (N). **[6]**

1. Write the full decay equation, including the antineutrino.
2. State how the nucleon number and proton number of the daughter compare with the parent, and confirm charge is conserved.

End of Section A — 150 structured questions.

SECTION B — Multiple choice [Q1–Q100] (choose one: A, B, C or D)

1 Physical quantities and units

1. The power P dissipated in a resistor is $P = kI^aR^b$, where I is current and R is resistance. For the equation to be homogeneous, a and b are:

A a = 1, b = 1
B a = 2, b = 1
C a = 2, b = 2
D a = 1, b = 2

---

2. A quantity is calculated from $Y = p^2q / \sqrt{r}$. The percentage uncertainties are p: 2%, q: 3%, r: 8%. The percentage uncertainty in Y is:

A 9%
B 11%
C 13%
D 17%

---

3. Which of the following is a set of three vector quantities only?

A mass, velocity, force
B displacement, acceleration, momentum
C work, power, energy
D velocity, speed, weight

---

4. The Young modulus has SI base units:

A kg m s^{-2}

B kg m^{-1} s^{-2}

C kg m^2 s^{-2}

D kg m^{-2} s^{-1}

---

5. A micrometer has a zero error of -0.03 mm (it reads -0.03 mm when closed). A wire measured with it gives a reading of 0.42 mm. The true diameter is:

A 0.39 mm

B 0.42 mm

C 0.45 mm

D 0.48 mm

---

6. Which estimate is closest to the order of magnitude of the kinetic energy of a 60 kg sprinter running at 10 m s^{-1}?

A 3×10^2 J

B 3×10^3 J

C 3×10^4 J

D 3×10^5 J

---

7. Two forces of 7.0 N and 4.0 N act at a point. Which value could NOT be the magnitude of their resultant?

A 3.0 N

B 7.0 N

C 11.0 N

D 2.0 N

---

8. A student states a result as $(4.0 \pm 0.2) \times 10^3$ and another quantity as $(2.00 \pm 0.05) \times 10^2$. The percentage uncertainty in their product is closest to:

A 2.5%

B 5.0%

C 7.5%

D 10%

---

9. Which pair of quantities has the same SI base units?

A energy and power

B pressure and stress

C momentum and force

D density and pressure

---

10. The equation for the speed of a wave on a string is $v = \sqrt{T/\mu}$, where T is tension. For homogeneity, the base units of μ must be:

A $kg\ m^{-1}$

B kg m

C $kg\ s^{-1}$

D $kg\ m^{-2}$

---

2 Kinematics

11. A ball is thrown vertically up and returns to the thrower. Taking up as positive, which graph best describes its acceleration against time during the whole flight (ignoring air resistance)?

- **A** a positive constant then negative constant
- **B** a constant negative value throughout
- **C** zero at the top, negative elsewhere
- **D** a straight line decreasing through zero

12. A car accelerates uniformly from rest. In the first second it travels distance d. The distance it travels in the third second is:

- **A** 3d
- **B** 5d
- **C** 7d
- **D** 9d

13. A projectile is launched at 30° to the horizontal with speed u. At the highest point of its path, its speed is:

- **A** 0
- **B** u sin30°
- **C** u cos30°
- **D** u

14. The displacement–time graph of an object is a curve that becomes steeper with time. This represents:

- **A** constant velocity
- **B** increasing velocity (acceleration)
- **C** decreasing velocity
- **D** the object at rest

15. Two stones are dropped from a cliff, the second 1.0 s after the first. As they fall (no air resistance), the vertical distance between them:

- **A** stays constant
- **B** increases
- **C** decreases
- **D** first increases then decreases

16. A ball falls from rest and passes a window 1.5 m tall in 0.20 s. Approximately how far above the top of the window was it released? ($g = 9.8\ m\ s^{-2}$)

A 1.5 m

B 2.0 m

C 2.5 m

D 3.0 m

---

17. An object moves with constant velocity in the x-direction and constant acceleration in the y-direction (starting with zero y-velocity). Its path is:

A a straight line

B a circle

C a parabola

D a hyperbola

---

18. A car travels 40 m at 20 $m\ s^{-1}$ then 40 m at 40 $m\ s^{-1}$. Its average speed over the 80 m is:

A 24 $m\ s^{-1}$

B 26.7 $m\ s^{-1}$

C 30 $m\ s^{-1}$

D 33.3 $m\ s^{-1}$

---

19. A velocity–time graph is a straight line from (0 s, 8 $m\ s^{-1}$) to (4 s, 0 $m\ s^{-1}$), then continues as a straight line to (6 s, -6 $m\ s^{-1}$). The total distance travelled (not displacement) in 6 s is:

A 10 m

B 16 m

C 22 m

D 6 m

---

20. A stone thrown horizontally from a height h lands a horizontal distance R away. If it were thrown horizontally at twice the speed from the same height, the horizontal distance would be:

A R

B $\sqrt{2}$ R

C 2R

D 4R

---

3 Dynamics

21. A 2.0 kg body moving at 3.0 m s^{-1} east experiences a constant force and, 4.0 s later, moves at 3.0 m s^{-1} west. The magnitude of the average force is:

A 0 N

B 1.5 N

C 3.0 N

D 6.0 N

---

22. A ball of mass m hits the ground vertically at speed v and rebounds at speed v. The magnitude of the impulse on the ball is:

A 0

B mv

C 2mv

D mv^2

---

23. Two trolleys, masses m and 2m, moving towards each other at the same speed v, collide and stick together. Their common velocity has magnitude:

A 0

B v/3

C v/2

D 2v/3

---

24. A resultant force–time graph for an object is a triangle: rising from 0 to F_0 over time τ then falling to 0 over the next τ. The change in momentum is:

A $F_0\tau$

B $\frac{1}{2}F_0\tau$

C $2F_0\tau$

D $F_0\tau/4$

---

25. An object falls through air and reaches terminal velocity. At terminal velocity:

A the drag force is zero

B the weight is zero

C the resultant force is zero

D the acceleration equals g

---

26. A 0.20 kg ball moving at 5.0 m s^{-1} is struck and moves off at 5.0 m s^{-1} at 90° to its original direction. The magnitude of the change in momentum is:

A 0

B 1.0 kg m s^{-1}

C 1.41 kg m s^{-1}

D 2.0 kg m s^{-1}

---

27. In an elastic collision between two equal masses, one initially at rest, the moving mass:

A stops and the other moves off with its velocity

B continues with the same velocity

C reverses direction

D and the target move off together

---

28. A rocket ejects 50 kg of gas per second at 800 m s^{-1} relative to the rocket. The thrust is:

A 1.6×10^{1} N

B 4.0×10^{4} N

C 4.0×10^{3} N

D 6.4×10^{5} N

---

29. A body of constant mass has a momentum–time graph that is a straight line of positive gradient. This tells us that:

A the velocity is constant

B the resultant force is constant and non-zero

C the body is in equilibrium

D the acceleration is zero

---

30. A person of weight W stands in a lift accelerating downward at a. The reading on scales beneath them is:

A W(1 + a/g)

B W(1 - a/g)

C W

D Wa/g

---

31. Two objects interact. Which statement is always true, whatever the type of collision?

A kinetic energy is conserved

B total momentum is conserved

C the objects have equal speeds afterwards

D the total kinetic energy increases

---

32. A trolley of mass 1.0 kg moving at 4.0 m s^{-1} collides with a stationary 3.0 kg trolley and they move off together. The fraction of the original kinetic energy that is lost is:

A 1/4

B 1/2

C 3/4

D 0

4 Forces, density and pressure

33. A uniform metre rule is pivoted at its centre. A 2.0 N weight hangs at the 20 cm mark. To balance it, a 4.0 N weight must hang at the:

- **A** 65 cm mark
- **B** 70 cm mark
- **C** 80 cm mark
- **D** 90 cm mark

34. The pressure at a depth of 20 m in seawater (density 1030 kg m^{-3}) due to the water alone is approximately:

- **A** 2.0×10^4 Pa
- **B** 1.0×10^5 Pa
- **C** 2.0×10^5 Pa
- **D** 2.0×10^6 Pa

35. A couple consists of two forces of 6.0 N separated by 0.25 m. Its torque is:

- **A** 0.75 N m
- **B** 1.5 N m
- **C** 3.0 N m
- **D** 0 N m

36. A block floats in water with 80% of its volume submerged. Its density is:

- **A** 200 kg m^{-3}
- **B** 800 kg m^{-3}
- **C** 1000 kg m^{-3}
- **D** 1250 kg m^{-3}

37. A ladder rests against a smooth wall on rough ground. The frictional force at the ground acts:

- **A** vertically up
- **B** vertically down
- **C** horizontally towards the wall
- **D** horizontally away from the wall

38. An object of volume V and density ρ_o is fully immersed in a fluid of density ρ_f (with $\rho_o > \rho_f$). The apparent weight (true weight minus upthrust) is:

A $\rho_f Vg$

B $(\rho_o - \rho_f)Vg$

C $(\rho_o + \rho_f)Vg$

D $\rho_o Vg$

---

39. Three forces act on a body in equilibrium. When drawn head-to-tail they must:

A all be parallel

B form a closed triangle

C form a right angle

D have equal magnitudes

---

40. A non-uniform beam of weight 200 N rests on two supports 3.0 m apart, one at each end. The left support reads 120 N and the right support reads 80 N. The distance of the centre of gravity from the left support is:

A 1.0 m

B 1.2 m

C 1.5 m

D 1.8 m

---

5 Work, energy and power

41. A 2.0 kg object is raised 5.0 m and also given a speed of 4.0 m s^{-1}. The total energy supplied (no losses) is:

A 16 J

B 98 J

C 114 J

D 196 J

---

42. A car of power output P travels at constant speed v against a resistive force F. Which is correct?

A $P = Fv$

B $P = F/v$

C $P = Fv^2$

D $P = \frac{1}{2}Fv$

---

43. The braking distance of a car on a level road is 20 m at 15 m s^{-1}. With the same braking force, the braking distance at 30 m s^{-1} is:

A 40 m

B 60 m

C 80 m

D 160 m

---

44. A pump raises 300 kg of water per minute to a height of 12 m. The minimum power required is approximately:

A 59 W

B 590 W

C 3500 W

D 35 W

---

45. A 60 kg skier descends a 20 m vertical drop and arrives with KE of 9.0 kJ. The energy dissipated by friction is approximately:

A 2.8 kJ

B 9.0 kJ

C 11.8 kJ

D 20.8 kJ

---

46. An object is projected vertically up with kinetic energy E. At the height where its speed has halved, its kinetic energy is:

A E/2

B E/4

C 3E/4

D E

---

47. A machine has an efficiency of 40%. To deliver 800 J of useful output, the energy input required is:

A 320 J

B 1120 J

C 2000 J

D 3200 J

---

48. A constant force of 20 N pulls a box 5.0 m across a floor; the force acts at 60° to the horizontal (direction of motion). The work done by the force is:

A 50 J

B 87 J

C 100 J

D 173 J

---

49. Water flows over a waterfall of height 50 m at 200 kg s^{-1}. If all the PE were converted to electrical energy, the maximum power output would be about:

A 9.8 kW

B 98 kW

C 980 kW

D 9.8 MW

---

6 Deformation of solids

50. A wire extends 2.0 mm under a load. A second wire of the same material and length but twice the diameter carries the same load. Its extension is:

A 0.5 mm
B 1.0 mm
C 2.0 mm
D 4.0 mm

---

51. The area under a force–extension graph (within the limit of proportionality) represents:

A the spring constant
B the stress
C the elastic potential energy stored
D the Young modulus

---

52. Two springs each of stiffness k are joined end-to-end (in series). The combined stiffness is:

A 2k
B k
C k/2
D k/4

---

53. A material is stretched beyond its elastic limit and then unloaded. On the force–extension graph, this shows as:

A the line returning exactly along the loading line
B a permanent extension when the force reaches zero
C an increased spring constant
D no extension at all

---

54. A wire of Young modulus E, length L and cross-sectional area A is stretched by force F. The extension is:

A FL/(AE)
B FA/(LE)
C EAL/F
D FE/(AL)

---

55. A steel wire and a copper wire have the same dimensions and carry the same load within their limits of proportionality. Steel has the larger Young modulus. Compared with the copper wire, the steel wire has:

- **A** greater stress and greater strain
- **B** the same stress but smaller strain
- **C** smaller stress and greater strain
- **D** the same stress and the same strain

---

56. A spring stores 0.80 J of elastic PE at an extension of 40 mm (within its limit of proportionality). At an extension of 20 mm it stores:

- **A** 0.20 J
- **B** 0.40 J
- **C** 0.60 J
- **D** 0.80 J

---

7 Waves

57. A wave has speed 340 m s^{-1} and frequency 170 Hz. Two points 1.5 m apart along the direction of travel have a phase difference of:

- **A** 90°
- **B** 180°
- **C** 270°
- **D** 360°

---

58. On a CRO the time-base is 5.0 ms/div and one cycle spans 4.0 divisions. The frequency is:

- **A** 25 Hz
- **B** 50 Hz
- **C** 100 Hz
- **D** 200 Hz

---

59. The intensity of a wave is quadrupled. Its amplitude changes by a factor of:

- **A** $\sqrt{2}$
- **B** 2
- **C** 4
- **D** 16

---

60. A source of sound moves towards a stationary observer at speed v_S; the speed of sound is v. The observed frequency is:

- **A** $f_S(v - v_S)/v$
- **B** $f_Sv/(v + v_S)$
- **C** $f_Sv/(v - v_S)$
- **D** $f_S(v + v_S)/v$

---

61. Which electromagnetic radiation has the shortest wavelength in free space?

- **A** microwaves
- **B** infrared
- **C** ultraviolet
- **D** gamma rays

---

62. Plane-polarised light of intensity I_0 passes through a polariser whose axis is at 30° to the plane of polarisation. The transmitted intensity is:

- **A** 0.25 I_0
- **B** 0.50 I_0
- **C** 0.75 I_0
- **D** 0.87 I_0

---

63. A sound wave passes from air into water, where its speed is greater. Which quantity is unchanged?

- **A** speed
- **B** wavelength
- **C** frequency
- **D** intensity

---

64. Which statement about a longitudinal wave is correct?

- **A** it can be polarised
- **B** particles oscillate perpendicular to travel
- **C** it consists of compressions and rarefactions
- **D** it cannot transfer energy

---

65. Two points on a transverse progressive wave are exactly half a wavelength apart. Their motions are:

- **A** in phase
- **B** in antiphase (180° out of phase)
- **C** 90° out of phase
- **D** always both at maximum displacement

---

66. Unpolarised light of intensity I_0 passes through one ideal polarising filter. The transmitted intensity is:

- **A** I_0
- **B** 0.71 I_0
- **C** 0.50 I_0
- **D** 0.25 I_0

---

8 Superposition

67. In a double-slit experiment, the fringe separation is x. If the slit separation is halved and the screen distance is doubled, the new fringe separation is:

- **A** x
- **B** 2x
- **C** 4x
- **D** x/4

---

68. A diffraction grating of 300 lines per mm is used with light of wavelength 500 nm. The number of diffracted orders visible on each side of the centre is:

- **A** 4
- **B** 5
- **C** 6
- **D** 7

---

69. Adjacent nodes on a stationary wave are 0.30 m apart. The wavelength of the waves forming it is:

- **A** 0.15 m
- **B** 0.30 m
- **C** 0.60 m
- **D** 1.2 m

---

70. For a stable two-source interference pattern, the two sources must be:

- **A** of different frequency
- **B** coherent (constant phase difference)
- **C** incoherent
- **D** of perpendicular polarisation

---

71. A stationary wave differs from a progressive wave in that a stationary wave:

- **A** transfers energy along its length
- **B** has all points oscillating with the same amplitude
- **C** does not transfer energy along its length
- **D** has no nodes

---

72. White light passes through a diffraction grating. In the first-order spectrum, the colour diffracted through the largest angle is:

- **A** violet
- **B** green
- **C** red
- **D** all diffracted equally

---

73. Two coherent sources produce interference. At a point the path difference is 2.5λ. This point is:

- **A** a bright fringe (constructive)
- **B** a dark fringe (destructive)
- **C** of intermediate intensity
- **D** outside the pattern

---

74. The most pronounced diffraction of a wave through a gap occurs when the gap width is:

- **A** much larger than the wavelength
- **B** approximately equal to the wavelength
- **C** much smaller than the wavelength
- **D** independent of the wavelength

---

9 Electricity

75. A current of 5.0 A flows for 4.0 minutes. The number of electrons passing a point is ($e = 1.6 \times 10^{-19}$ C):

- **A** 7.5×10^{21}
- **B** 1.9×10^{21}
- **C** 7.5×10^{19}
- **D** 1.2×10^{3}

---

76. For a wire carrying current I with n charge carriers per unit volume, each of charge q, and cross-sectional area A, the drift speed is:

- **A** I/(nAq)
- **B** nAq/I
- **C** InAq
- **D** IA/(nq)

---

77. A wire is stretched to twice its length (volume constant). Its resistance becomes:

- **A** half
- **B** double
- **C** four times
- **D** unchanged

---

78. The I–V graph of a filament lamp is a curve that bends towards the V-axis as V increases. This is because:

- **A** the filament cools with current
- **B** the resistance decreases with current
- **C** the resistance increases as temperature rises
- **D** it obeys Ohm's law

---

79. Two resistors 3.0 Ω and 6.0 Ω are in parallel. The combined resistance is:

- **A** 9.0 Ω
- **B** 4.5 Ω
- **C** 2.0 Ω
- **D** 0.5 Ω

---

80. The resistance of a thermistor (NTC) as temperature rises, and of an LDR as light intensity rises, respectively:

- **A** increases; increases
- **B** decreases; decreases
- **C** increases; decreases
- **D** decreases; increases

---

81. A 12 V, 36 W lamp operates at its rating. Its resistance is:

- **A** 0.33 Ω
- **B** 3.0 Ω
- **C** 4.0 Ω
- **D** 48 Ω

---

82. Power P is transmitted at voltage V through cables of resistance R. The power lost as heat in the cables is:

- **A** P^2R/V^2
- **B** PV/R
- **C** V^2/R
- **D** PR/V

---

83. Two identical resistors are connected first in series, then in parallel, across the same supply. The ratio (power in parallel : power in series) is:

- **A** 1 : 1
- **B** 2 : 1
- **C** 4 : 1
- **D** 1 : 4

---

10 D.C. circuits

84. A cell of e.m.f. 1.5 V has internal resistance 0.50 Ω. When connected to a 1.0 Ω resistor, the terminal p.d. is:

- **A** 0.50 V
- **B** 1.0 V
- **C** 1.25 V
- **D** 1.5 V

---

85. Kirchhoff's first law (current) is a consequence of the conservation of:

- **A** energy
- **B** charge
- **C** momentum
- **D** mass

---

86. In a potential divider, two equal resistors are across a 10 V supply. The output across one resistor is 5.0 V. If a load equal to R is connected across that resistor, the output becomes:

- **A** 5.0 V
- **B** 3.3 V
- **C** 2.5 V
- **D** 6.7 V

---

87. A battery delivers maximum power to an external resistor R when:

- **A** R is very large
- **B** R = 0
- **C** R equals the internal resistance r
- **D** R is very small

---

88. The e.m.f. of a source is best defined as the energy transferred per unit charge:

- **A** dissipated in the internal resistance
- **B** in driving charge around the complete circuit
- **C** in the external circuit only
- **D** when no charge flows

---

89. Three 6.0 Ω resistors are connected in parallel. The combined resistance is:

- **A** 18 Ω
- **B** 6.0 Ω
- **C** 3.0 Ω
- **D** 2.0 Ω

---

90. A potentiometer is balanced (galvanometer reads zero) at length 45.0 cm for one cell and 60.0 cm for a second. The ratio of their e.m.f.s (first : second) is:

- **A** 3 : 4
- **B** 4 : 3
- **C** 1 : 1
- **D** 9 : 16

---

91. As the current drawn from a real battery increases, its terminal potential difference:

- **A** increases
- **B** decreases
- **C** stays equal to the e.m.f.
- **D** becomes zero immediately

---

92. In the circuit, a 6.0 V battery (negligible internal resistance) is connected to a 2.0 Ω resistor in series with a parallel pair of 4.0 Ω resistors. The current from the battery is:

- **A** 1.0 A
- **B** 1.5 A
- **C** 2.0 A
- **D** 3.0 A

---

11 Particle physics

93. The nuclide $^{238}_{92}U$ emits an α-particle. The resulting nuclide has:

A A = 234, Z = 90

B A = 236, Z = 90

C A = 234, Z = 92

D A = 238, Z = 90

94. During β^- decay, within the nucleus:

A a proton becomes a neutron

B a neutron becomes a proton

C an up quark becomes a down quark

D a neutron becomes a positron

95. The charge on a neutron (udd) confirms which quark charges?

A u = +⅓, d = -⅔

B u = +⅔, d = -⅓

C u = -⅔, d = +⅓

D u = +1, d = 0

96. A meson consists of:

A three quarks

B two quarks

C one quark and one antiquark

D a lepton and a quark

97. β-particles are emitted with a continuous range of energies because:

A the nucleus recoils

B a neutrino/antineutrino shares the energy

C γ-rays are also emitted

D the electrons collide with atoms

98. Which particle is a lepton?

A proton

B neutron

C neutrino

D pion (meson)

---

99. In the α-scattering experiment, the small fraction of α-particles deflected through large angles indicates that the atom's positive charge is:

A spread uniformly through the atom

B concentrated in a tiny nucleus

C carried by the electrons

D zero

---

100. Which quantities are always conserved in a nuclear decay equation?

A nucleon number and charge

B mass and kinetic energy

C proton number and neutron number separately

D number of α-particles

End of Section B — 100 multiple-choice questions.